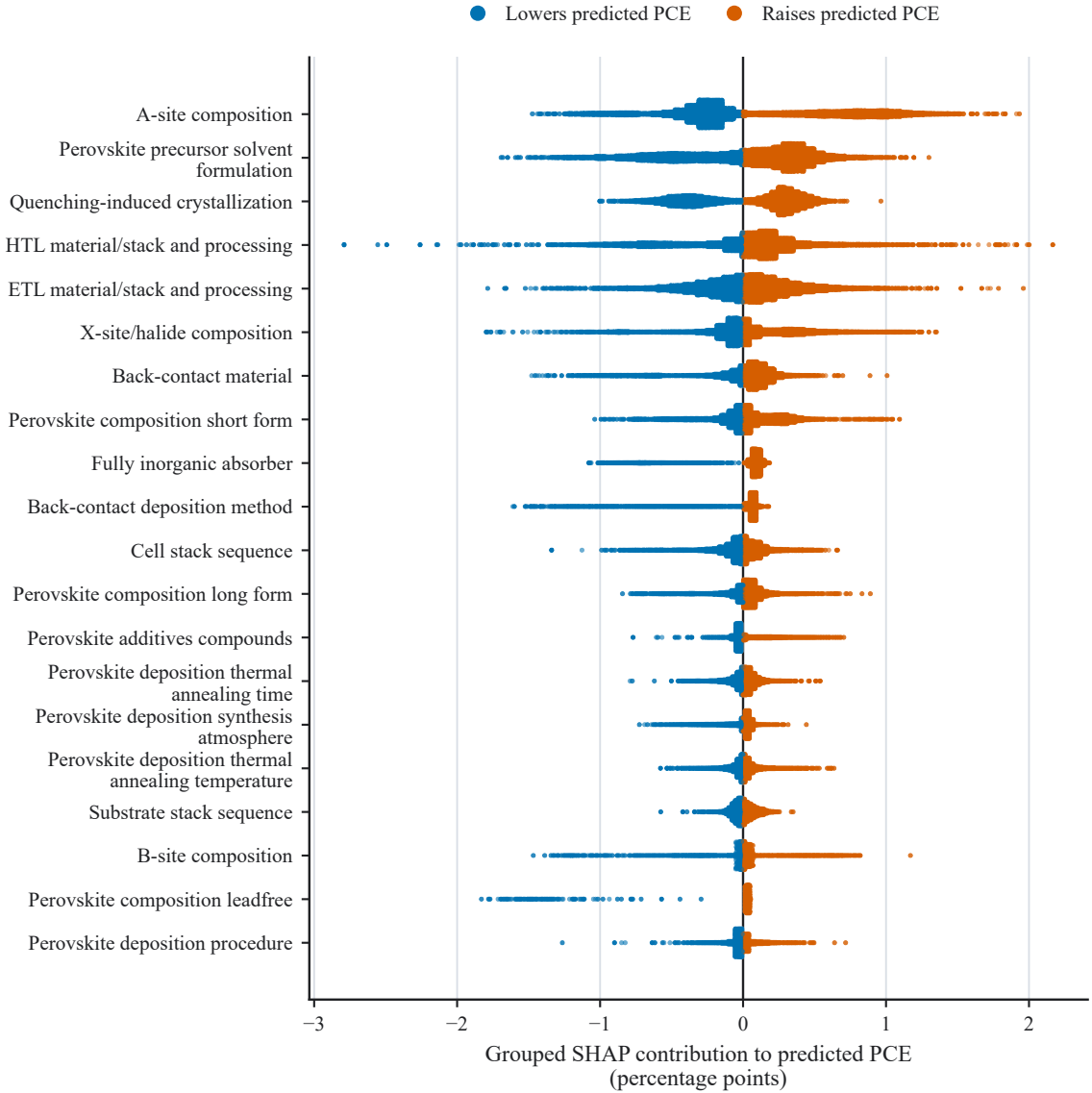

Lowers predicted PCE
Raises predicted PCE
A-site composition
Perovskite precursor solvent formulation
Quenching-induced crystallization
HTL material/stack and processing
ETL material/stack and processing
X-site/halide composition
Back-contact material
Perovskite composition short form
Fully inorganic absorber
Back-contact deposition method
Cell stack sequence
Perovskite composition long form
Perovskite additives compounds
Perovskite deposition thermal annealing time
Perovskite deposition synthesis atmosphere
Perovskite deposition thermal annealing temperature
Substrate stack sequence
B-site composition
Perovskite composition leadfree
Perovskite deposition procedure
−3
−2
−1
0
1
2
Grouped SHAP contribution to predicted PCE (percentage points)